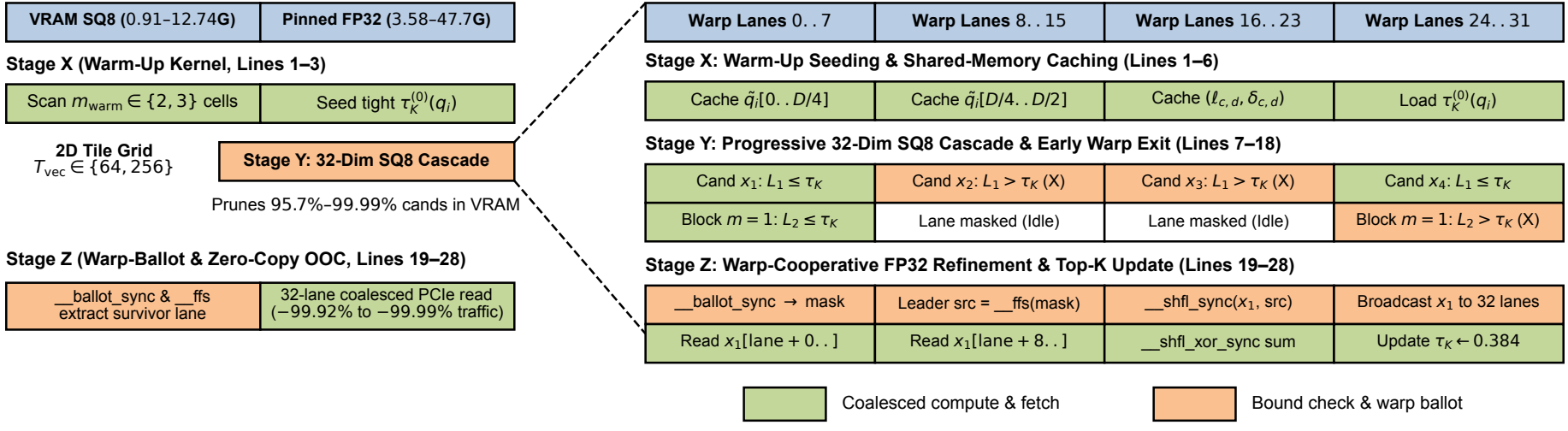



**(a) Three-stage pipeline & zero-copy OOC**

**(b) Spatiotemporal warp execution across Stages X–Z**